



CERILLION TECHNOLOGIES

CWS

Cerillion Operations Training

Module 12 - Billing Process

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Module Introduction

Overview

This module introduces the key concepts on the Billing Processes and the Bill Cycle Maintenance aspect of the Cerillion CRM. The purpose of this document is to provide an overview of the related processes in Cerillion in these areas.

By the end of this module, you should:

- Understand the steps required to run 'Billing'.
- Understand Bill Generation, Bill Printing and Bill Formatting.

Pre-requisites

It is assumed that you are familiar with Account, Service, Equipment and Work Order concepts in Cerillion.

It is assumed that you are familiar with Rentals, Usage and other adhoc charges and their respective transaction codes.

Basic idea of summarization process in Cerillion.

1. Pre Billing Checks for Real Billing

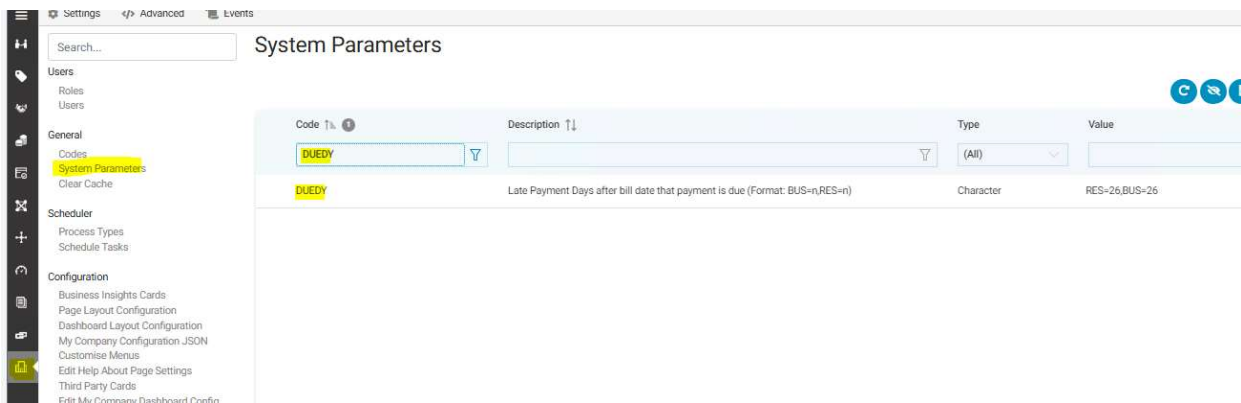
- Change Due Date parameter as per when you start billing.

For eg – If real billing started on 1st March and due date is 30th March then we need to set DUEY parameter value as RES=29,BUS=29

If Real billing started on 5th March and due date is 30th March then we need to set DUEY parameter value as RES=25,BUS=25

To set DUEY parameter value goto CRM → My Company → System Parameters

Search for DUEY and update the value.



- Move third party payer accounts to A1A or A1U cycle

Run the query below to check if any payer account exists on bill cycle other than A1A & A1U.

```
select s.bill_cycle_code, s.account_no, s.creation_source_cd,
sp.account_no payer_account_no, sp.account_name1 payer_account_name
, s.creation_source_cd payer_account_type, sp.bill_cycle_code
payer_bill_cycle_code, tpp.*
from subscribers s, subscribers sp, third_party_payments tpp,
equipment_charging ec
where sp.account_no = tpp.third_account_no
and tpp.prime_uid = ec.third_payment_uid
and ec.account_no = s.account_no
and ( tpp.end_dt is null or tpp.end_dt >= '01-May-2024')
and sp.bill_cycle_code not in ('A1A','A1U');
```

Update the bill cycle code of payer account to A1A or A1U depending on which currency they belong.

A1A for Seychelles currency and A1U for USD

- In Bill cycle parameters all below bill cycle code has expected next_bill_date and last_st is 0.

M1U, M1V, M1C, M1G, M1F, M1B, M1A, M1R, A1A, A1U

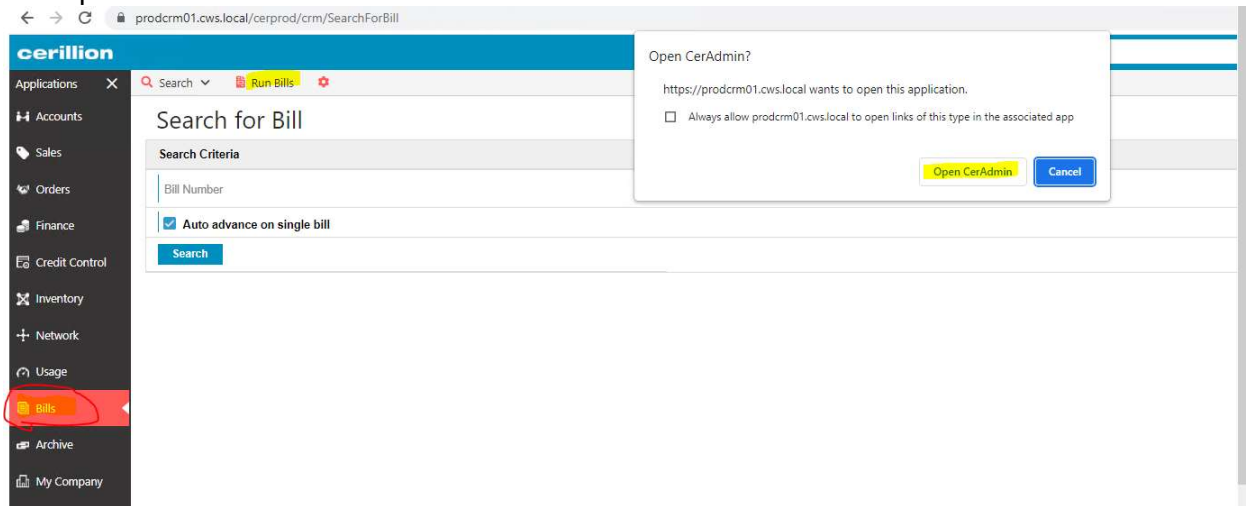
- Bill cycle A1A & A1U contain payer accounts so, we need to run billing for these 2 cycles at end when billing of other cycles completed successfully.
- Ensure new bill images are uploaded.
- Follow below sequence of steps
 - Pre billrun checks
 - Start Test/Real billing from Backend or using CRM except A1A & A1U cycles
 - Validate bill run logs
 - If all cycles completed successfully start bill run for A1A & A1U
 - Validate bill run logs
 - Start bill print for all completed cycles. (Bill print can be started anytime once bill run of cycle completed)
 - Check if any account failed in bill print
 - Verify PDF generation
 - Post bill check, verify all PDF generated successfully

2. CRM Billing Process

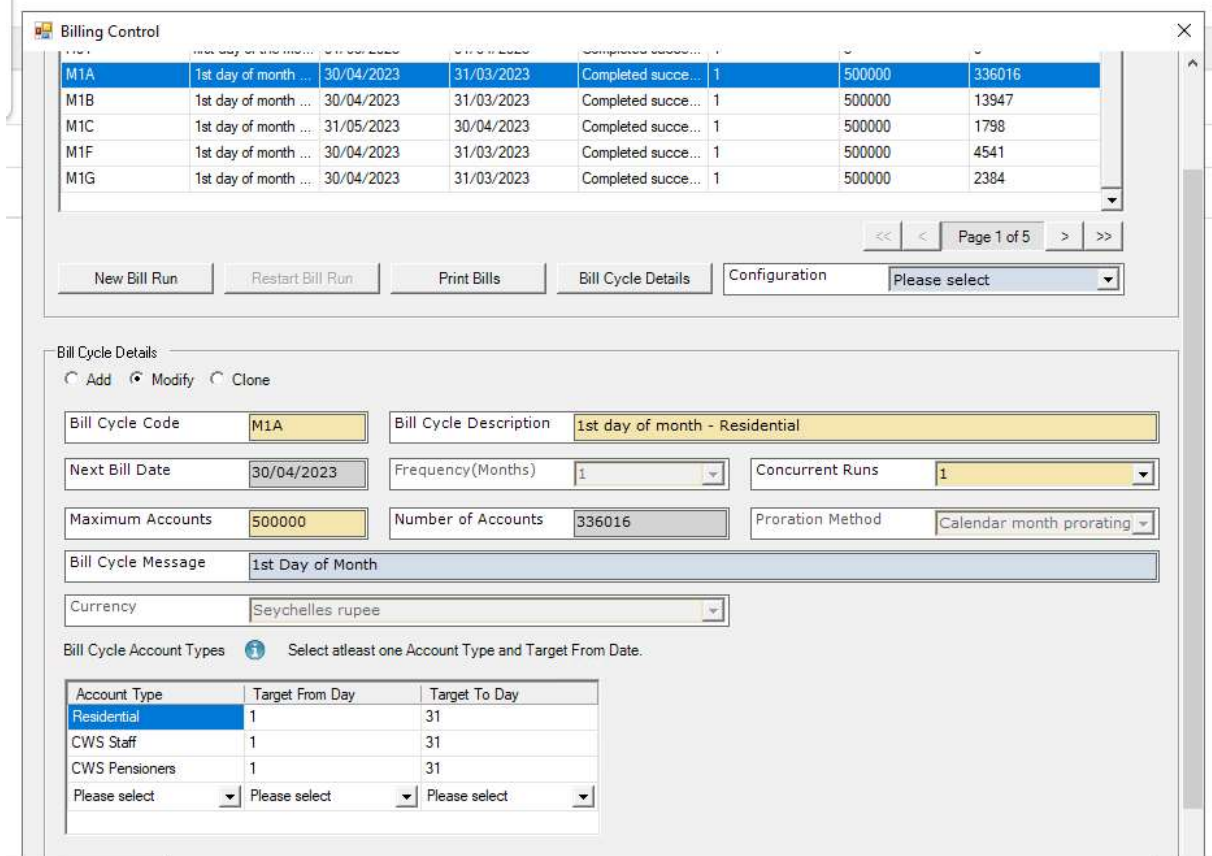
Billing can be started from CRM screen as well as from backend.

2.1 Run Billing from Screen

The Billing Control screen is available via the thick admin client accessed from the Bills / Run Bills menu options within the CRM.



Below is a screenshot of the Billing Control screen.



The Bill Cycles section at the top of the screen displays all the different billing cycles and provides additional information on each cycle such as its billing description, last run date, next run date, last bill run status etc.

The <Bill Cycle Details> fields allow for additional control of the bill cycle, for e.g. to specify the number of accounts or to add a message to the bill cycle.

2.2 New Bill Run – Bill Generation Process

In order to execute a bill run, the following steps need to be followed:

1. Go to the Thick Client Administrator (i.e., the Billing Control screen)
2. In Billing Control / Bill Cycles section, select an entry in the Bill Cycle Code column. This is the bill cycle for which you want to execute billing.
3. Select the <New Bill Run> button. This enables you to modify the details of the bill run as explained below:
 - a. Bill Run Type – you can choose to run billing in Test mode or in Production mode. Test billing does not commit transactions to the database and hence you can assess a bill run before running it in production.
 - b. Collection Plan Start Date – choose the date based on when the cycle is executed. This allows for flexibility should a bill run be delayed.
 - c. Payment Due Date – change the default value if required for the current instance of the bill run.
 - d. Grace Period – change the default value if required for the current instance of the bill run.
4. Select the <Start Bill Run> button in the Bill Printing Details section once you are ready to execute the bill run. See the screenshot below.

The screenshot shows the 'Billing Control' application window. The 'Bill Cycles' section contains a table with the following data:

Bill Cycle Code	Bill Cycle Description	Next Bill Date	Last Run Date	Last Run Status	Frequency(Months)	Maximum Accounts	Number of Accounts
DEM	Demand Bill Cycle	01/05/2023	30/11/1999	Completed succe...			
H01	first day of the mo...	01/04/2023	01/04/2023	Completed succe...	1	3	0
M1A	1st day of month ...	30/04/2023	31/03/2023	Completed succe...	1	500000	336016
M1B	1st day of month ...	30/04/2023	31/03/2023	Completed succe...	1	500000	13947
M1C	1st day of month ...	31/05/2023	30/04/2023	Completed succe...	1	500000	1798
M1F	1st day of month ...	30/04/2023	31/03/2023	Completed succe...	1	500000	4541
M1G	1st day of month ...	30/04/2023	31/03/2023	Completed succe...	1	500000	2384

Below the table are buttons: 'New Bill Run' (highlighted in yellow), 'Restart Bill Run', 'Print Bills', 'Bill Cycle Details', 'Configuration', and a 'Please select' dropdown. The 'New Bill Run Details' section contains the following fields:

- Bill Run Type: Production (selected)
- Bill Cycle Code: Production (selected)
- Collection Plan Start Date: 06/07/2023
- Previous Bill Date: 31/03/2023
- Late Payment Charge Days: 27
- Late Payment Grace Period (days): 2
- Bill To Date: 30/04/2023

A 'Start Bill Run' button is highlighted in yellow at the bottom of the 'New Bill Run Details' section.

5. Bill runs can be monitored in the backend. Login to the Linux server for your environment and navigate to the 'log' directory.

```
cd /cer_<ENV Name>/log
```

For e.g. on CERPROD this would be:

```
/cer_cerprod/log
```

The billing run log file will have a naming convention of:

bil0101b_MMDDHHMMSS00_XXXXX.log where HHMMSS is the time when the process was started, (e.g., bil0101b_070619203800_2104402.log).

2.3 Billing reports

Once billing completed below report get generated. For test bill report will be generated in path "/cer_cerprod/work/bil/test" and for real bill report generated in path "/cer_cerprod/work/bil"

BIL0105R_121310542174.REP □ Items Not Billed Report

BIL0103R_121310542176.REP □ Billing Transactions Report

BIL0102R_121310542175.REP □ Bill Summary Report

See below screenshot:

□ Bill Summary Report

```
cwspd@prodbe01>cat BIL0102R_070619210255.REP
```

ABS - Billing		Bill Summary Report		06/07/23 19:21	
Page 1					

Report Date: 06/07/23 19:21 Bill Cycle Code: MIC Currency: SCR Bill To Date: 01/05/23 Run No: 2002Calling Use					
r: SUPER					
No. Accounts Due	Total Amount Due	No. Accounts Not Billed	Total Amount Not Billed	No. Accounts Billed	Total Amount Billed
1204	104214.61	539	0.00	665	104214.61

Subscriber Type	Highest Bill Amount	Lowest Bill Amount	Average Bill Amount		
BC	0.00	0.00	0.00		
CP	0.00	0.00	0.00		
CW	13149.60	0.00	102.00		
PA	100.00	100.00	100.00		
PM	33827.44	0.00	732.47		
RE	1280.23	803.85	1042.04		
ST	3522.30	-1495.00	1085.36		

Subscriber Type	Transaction Type	Highest Billed Charge	Lowest Billed Charge	Average Billed Charge	Income Generated
BC	CHRENT	0.00	0.00	0.00	0.00
CP	CHRENT	0.00	0.00	0.00	0.00
CW	CH13CA	31.60	0.40	10.37	51.87
	CH13IA	75.83	70.00	72.92	145.83
	CH13TA	32.96	30.42	31.69	63.38
	CH14CA	130.00	130.00	130.00	130.00

□ Billing Transactions Report

```
cwspdp@prodbe01>
cwspdp@prodbe01>cat BIL0103R_070619210256.REP
```

ABS - Billing e 1	Daybook of Billing Transactions Report	06/07/23 19:21	Page
----------------------	--	----------------	------

Report Date: 06/07/23 19:21	Bill Cycle Code: M1C	Currency: SCR	Bill To Date: 01/05/23	Run No: 2002	Calling User: SUPER
-----------------------------	----------------------	---------------	------------------------	--------------	---------------------

Transaction Type	Service Type	Total Account No	Sum Trans. Amount
CHRBRA		1	1421.87
CHRENT	BB	271	11531.96
CHRENT	GM	282	40599.93
CHRENT	LC	86	0.00
CHRENT	PW	1	0.00
CHRENT	TP	471	43835.56
CHRENT	TV	42	10845.12
CHRENT	XX	1	175.95
CRRA01		1	-2242.50
CRRA01	BB	1	-1121.25
CRRBCA		3	-1421.87

□ Items Not Billed Report

ABS - Billing Page 2	Items Not Billed Report	06/07/23 19:21
-------------------------	-------------------------	----------------

Report Date: 06/07/23 19:21	Bill Cycle Code: M1C	Currency: SCR	Bill To Date: 01/05/23	Run No: 2002	Calling User: SUPER
-----------------------------	----------------------	---------------	------------------------	--------------	---------------------

Category Code	Category Description	Account Status	Account No.	Current Plan Code	Inac Amount	Ac
B	No periodic and usage charges	SN	10173302	STFF		
B	No periodic and usage charges	SN	10173308	STFF		
B	No periodic and usage charges	SN	10174607	STFF		
B	No periodic and usage charges	SN	10174629	STFF		
B	No periodic and usage charges	SN	10177459	STFF		
B	No periodic and usage charges	SN	10177463	STFF		
B	No periodic and usage charges	SN	10179325	STFF		
B	No periodic and usage charges	SN	10182109	STFF		
B	No periodic and usage charges	SN	10182110	STFF		
B	No periodic and usage charges	SN	10182665	STFF		
B	No periodic and usage charges	SN	10183375	STFF		
B	No periodic and usage charges	SN	10183376	STFF		
B	No periodic and usage charges	SN	10183684	STFF		
B	No periodic and usage charges	SN	10183823	SBSD		
B	No periodic and usage charges	SN	10183830	STFF		

2.4 Bill Printing Process

Once the <New Bill Run> step has completed successfully, the bill print generation process can be started by selecting <Print Bills> on the Billing Control screen.

Bill Cycles

Bill Cycle Code	Bill Cycle Description	Next Bill Date	Last Run Date	Last Run Status	Frequency(Months)	Maximum Accounts	Number of Accounts
DEM	Demand Bill Cycle		30/11/1999	Completed succe...			
H01	first day of the mo...	01/05/2023	01/04/2023	Completed succe...	1	3	0
M1A	1st day of month ...	30/04/2023	31/03/2023	Completed succe...	1	500000	336016
M1B	1st day of month ...	30/04/2023	31/03/2023	Completed succe...	1	500000	13947
M1C	1st day of month ...	31/05/2023	30/04/2023	Completed succe...	1	500000	1798
M1F	1st day of month ...	30/04/2023	31/03/2023	Completed succe...	1	500000	4541
M1G	1st day of month ...	30/04/2023	31/03/2023	Completed succe...	1	500000	2384

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New Bill Run Restart Bill Run Print Bills Bill Cycle Details Configuration Please select

Bill Printing Details

Bill Run Type Production

Bill Cycle Code M1C Bill Due Date 07/07/2023 Bill Runs by Date 30/04/2023

From Account No To Account No

Start Printing

Itemisation ☒

Cerillion Administrator

Bill print process has been initiated.

OK

Again, the option in <Bill Run Type> to select the type of bill run (i.e., whether it is in Production or Testing environment) is available. In addition, the ability to modify other parameters such as bill due date, bill cycle date and range of account numbers is enabled.

The Production outputs from the <Print Bills> step can be found in:

/cer_[ENV NAME]/work/bil/

On CERPROD this would be:

/cer_cerprod/work/bil/

The Testing outputs from the <Print Bills> step can be found in:

/cer_[ENV NAME]/work/bil/test/ On

CERPROD this would be:

/cer_cerprod/work/bil/test/

The system is configured to batch bills up to a maximum number per file so many files are created per bill run. The naming convention of these print bill files is:

[BillCycleCode]_[YYYYDD]_nnnnn.DAT (nnnnn is a sequential number starting from 00001)

Examples of bill print file names are:

M1C_230430_00001.DAT

M1C_230430_00002.DAT

:

:

M1C_230430_00009.DAT

2.5 Bill XMLs

Once the bill print XMLs (i.e.DAT files) have been created, the XML files will be transferred to a configured folder for the Streamserve application.

Script DAT_filetransfer.sh scheduled in crontab it transfers DAT files to streamserve input path.

Once the .DAT files are placed in the input folder, they will be processed by Streamserve and produce PDF files of the invoices in the output folder.

An example of the input folder could be:

/cer_cerprod/streams/billxml_in

The output folder will be one that is accessible to CSRs, Billing and other users of the CRM who need visibility of the bills.

2.6 Bill XMLs

To view PDF from CRM screen we need to run workaround script.

docum_attachment_workaround.sh script present in exe directory. We need to run this script with parameter bill_dt once real billing pdf generation is completed. This script creates entry in docum_acttachment table. Then we can see all bill pdf from CRM.

Goto exe path and run script

./docum_attachment_workaround.sh **<bill_date>**

Bill date format - 01-MAR-2025

2.7 Test Billing

New Bill Run and Print Bill options can both be run in 'test' mode. 'Test' mode works exactly like 'Production' mode, but no information is added to the super schema in the database.

The intention of using Test billing would be to run it before Production billing to assess if bills are correct.

Test billing can be run and re-run without updating the database. Whereas Production bills can only be run once per bill period.

3. Backend Billing Process

3.1 New Bill Run – Billing Generation Process

To execute a bill run from the backend, the following steps need to be followed:

1. Go to a Linux Machine and log into the backend of the environment you will be using
2. Go to /cer_[ENV NAME]/exe
3. Type the following command with the necessary parameters changed:

```
P1='M1A' P2='T' P3='2025_03_01 00:00:00' P4='28' P5='2' P6="" P7="" P8=""
/cer_cerprod/exe/pspbil0101b.sh
```

P1: Bill Cycle Code

P2: Test or Real Bill T/N

P3: Collection Plan start date

P4: Late payment charge days

P5: Grace Period

P6: Not Mandatory

P7: Not Mandatory

P8: Not Mandatory

3.2 Bill Printing Process

To execute a bill print from the backend, the following steps need to be followed:

1. Go to a Linux Machine and log into the backend of the environment you will be using
2. Go to /cer_[ENV NAME]/exe
3. Type the following command with the necessary parameters changed:

```
P1='S' P2='PBCC=M1A|PITM=Y|PTEST=Y' P3='2025_03_01 00:00:00'
P4='2025_03_01 00:00:00' P5='N' P6='2002' P7='0' P8='99999999'
/cer_cerprod/exe/bil0705s.sh
```

P1: Operation

P2: PBCC = [Bill cycle code]

PITM = [Itemisation flag 'Y' / 'N']

PTEST = [Test Billing 'Y' / 'N']

P3: Bill to date

P4: Bill due date

P5: Disconnection ['Y' / 'N']

P6: Billing run uid from bill_history table in db (called prime_uid of billing_runs) For bill printing of test bill cycle take billing_run_uid from test schema.

P7: Start account number

P8: End account number

This will generate the DAT XML files. To find these files, follow the same instructions as Section 1.4 outlines.

3.3 Logs

Logs are used to verify a bill cycle has been successfully executed and bills have been generated.

Bill run log file name start with bil0101b

Bill Print log file name start with bil0705s

We need to redirect logs when executing bill run and bill print from backend unix server.

3.4 Queries and Commands to Check in a Bill Run

- Check for fatal errors in log file:
`grep -i fatal bil0101b_070619203800_2104402.log`
- Check billing has completed / failed from log :
`grep "Billing finished" bil0101b_070619203800_2104402.log`
- Check last_st=0 in bill_cycle_parameters table. (0 indicates successful. Any other value means there is an issue in the bill run.)
`select * from BILL_CYCLE_PARAMETERS where bill_cycle_code='M1C'; -- change bill cycle code`
- After bill run, to check the number of accounts billed, run below query:
`select * from billing_runs where bill_cycle_code='M1C';-- change bill_cycle_code.`
- Check for latest bill. Prime_uid is used in below sql.
`select * from bill_history where billing_run_uid=2002;`
-- This will show the number of accounts billed in a particular cycle.

Module Exercise

Your course instructor will provide you with a username and password for Cerillion CRM. Log into Cerillion and then:

- Create a new bill cycle, assign a few accounts to the same bill cycle and run billing, bill print for the same bill cycle.
- Run test billing and bill print followed by generation of bill pdf for few accounts.